A fixed coupon does not fall with market rates

Suppose our CFO has USD 1,000 of debt paying 5% annually, with interest every six months. Each payment is USD 25. Even if market rates fall, the lender is still entitled to the contractual 5%. The company cannot simply substitute the new market rate for the rate written in its bond.

A floating rate note, or FRN, works differently. Its coupon resets using a reference rate plus a contractual spread. If the reference rate falls, the next applicable coupon falls; if it rises, the coupon rises. The spread is a separate part of the contract and stays fixed in our examples.

Feature	Fixed-rate bond	Floating rate note
Coupon rule	Contractual fixed rate	Benchmark + spread
Rate changes	Do not change the coupon	Change future reset coupons
Example	5% annually	SOFR + 3% annually

$$\text{Coupon}_t = N \times \frac{\text{SOFR}_{t-0.5} + s}{2}$$

Here N is face value and s is the annual spread. If SOFR is 3% and the spread is 3%, a USD 1,000 bond pays USD 30 for the semester: 1,000 x (3% + 3%) / 2. Divide the annual coupon rate by two, not the spread alone.

What does the benchmark represent?

SOFR is a USD reference rate. Other markets use benchmarks such as SONIA in sterling, euro STR in euros and TONA in yen. The benchmark is not the issuer's credit spread. A borrower can face a lower benchmark while still having an expensive overall financing rate.

Our timing convention
For the calculations here, an illustrative six-month SOFR-based rate is observed at the start of the semester and paid at the end. Actual overnight SOFR contracts often compound daily rates in arrears. We are isolating the reset logic, not reproducing those settlement details.

Reset first; pay at the end of the semester

Consider a two-year FRN with USD 1,000 face value, semiannual payments and a coupon of SOFR + 3%. At t=0, SOFR is 3%, which fixes the coupon paid at t=0.5. The 4% observed at t=0.5 determines the next coupon, paid at t=1.0.



Payment date	SOFR used	Coupon rate	Interest paid
t=0.5	3% at t=0	6%	$30
t=1.0	4% at t=0.5	7%	$35
t=1.5	5% at t=1.0	8%	$40
t=2.0	6% at t=1.5	9%	$45

Total interest is USD 150. At t=2.0, the borrower also repays USD 1,000 principal, so the final bond cash outflow is USD 1,045. A new 7% SOFR observation at t=2.0 does not create another coupon: the bond has matured.

Reverse the path and the direction reverses

With SOFR observations of 7%, 6%, 5%, 4% and 3% at the same dates, the four coupons are USD 50, 45, 40 and 35. The last 3% is again unused. Lower reset rates reduce the issuer's interest expense, but they also reduce the investor's interest income.

Read the date before using the number
For a payment at t, use SOFR from t - 0.5. The newly observed rate on the payment date belongs to the following semester. The direction of the cash flow depends on whether we are looking at the borrower or the investor.

A swap changes exposure without providing a loan

An interest rate swap exchanges interest payments calculated on an agreed notional amount. In our single-currency example, the notional is not exchanged. It tells us how large the interest payments are; it is not a loan advanced by the counterparty.

Firm A agrees to pay floating and receive 5% fixed for two years on USD 1,000, with semiannual payments. Its counterparty has the opposite position. The fixed leg remains USD 25 each semester, while the floating leg changes with the applicable SOFR reset.

$$\text{Paid}_t = N\frac{\text{SOFR}_{t-0.5}}{2}, \qquad \text{Received}_t = N\frac{5\%}{2}$$

$$\text{Net cash flow}_t = \text{Received}_t - \text{Paid}_t$$

Work through one payment

If the rate set at the beginning of the semester is 4%, A pays USD 20 and receives USD 25. Its net cash flow is +USD 5: it receives money. The counterparty's net cash flow is -USD 5. Both cannot have a positive net cash flow on the same bilateral exchange.

SOFR used	A pays floating	A receives fixed	A net cash flow
3%	$15	$25	+$10
5%	$25	$25	$0
7%	$35	$25	-$10

Cash flow and cost have opposite signs
Throughout these notes, net cash flow means received minus paid. A label such as net swap cost means paid minus received, so its sign is reversed. At 4% SOFR, +USD 5 of net cash flow is the same economic outcome as -USD 5 of net swap cost. Neither is the swap's market value.

Follow the full path, then add the underlying debt

Keep A's receive-5%-fixed/pay-SOFR swap. SOFR observed at t=0, 0.5, 1.0, 1.5 and 2.0 is 3%, 4%, 5%, 6% and 7%. The payment schedule uses the first four observations.

Payment	Floating paid	Fixed received	Net cash flow
t=0.5	$15	$25	+$10
t=1.0	$20	$25	+$5
t=1.5	$25	$25	$0
t=2.0	$30	$25	-$5
Total	$90	$100	+$10

A receives USD 10 net over the two years. With a different SOFR path of 3%, 3.4%, 3.2%, 3.1% and 3%, floating payments are USD 15, 17, 16 and 15.50. Fixed receipts still total USD 100, so A receives USD 36.50 net. These totals add cash flows; they do not discount them or measure market value.

Return to the CFO with 5% fixed debt

Suppose this same company also pays 5% on its original USD 1,000 debt. The USD 25 fixed debt payment is offset by the USD 25 fixed swap receipt. What remains is the floating swap payment.

$$\text{Final annual cost} = 5\% + \text{SOFR} - 5\% = \text{SOFR}$$

The company can now benefit if SOFR falls, but it pays more if SOFR rises. The debt itself has not been rewritten. The company added a separate contract that changes the combined exposure.

Three checks before calculating
Identify whose cash flows we are measuring, use the start-of-period rate, and keep the debt separate from the swap until the final step. A swap receipt is not automatically a financing gain; evaluate it together with the exposure the CFO intended to change.